

Question 1: Basic Querying

Assignment

Write a SQL script to return the total number and percentage of ads per category which receive leads within a period of 24 hours in May 2017.

Interpretation

"receive leads within a period of 24 hours in May 2017"

I count ads that were posted in May 2017, and check whether each ad received at least one lead within 24 hours after it was posted.

Assumptions

1. Snowflake SQL syntax.
2. Multiple lead events for the same ad count as one ad with leads.

Query

```
with ads as (  
    select  
        seller_id,  
        ad_id,  
        category,  
        -- Combine the date and time fields into one timestamp.  
        timestamp_from_parts(ad_start_date, ad_start_time) as  
ad_start_ts  
    from ad_table  
    where ad_start_date >= '2017-05-01'  
        and ad_start_date < '2017-06-01'  
)  
  
lead_events as (  
    select  
        clsfd_seller_id as seller_id,  
        clsfd_ad_id as ad_id,  
        buyer_id,  
        -- Combine the date and time fields into one timestamp.  
        timestamp_from_parts(event_table.date,  
event_table.timestamp) as event_ts  
    from event_table  
    -- Lead event types.  
    where eventtype in ('PHONE_CLICK', 'EMAIL', 'BID',  
'URL_CLICK')  
)
```

```

ad_level as (
    select
        ads.category,
        ads.seller_id,
        ads.ad_id,
        count(lead_events.ad_id) > 0 as had_lead_within_24h
    from ads
    left join lead_events
        on lead_events.ad_id = ads.ad_id
        -- Match on seller as well, in case ad_id is not unique
    across sellers.
        and lead_events.seller_id = ads.seller_id
        and lead_events.event_ts between ads.ad_start_ts and
ads.ad_start_ts + interval '24 hours'
    group by
        ads.category,
        ads.seller_id,
        ads.ad_id
)

select
    category,
    count(*) as total_ads,
    count_if(had_lead_within_24h) as ads_with_leads_24h,
    count_if(had_lead_within_24h) / count(*) as
share_ads_with_leads_24h
from ad_level
group by category
order by share_ads_with_leads_24h desc;

```